

CHE 321 HW 10

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1)

$$E_A = 100 \frac{\text{kJ}}{\text{mol}}$$

$$T = 400\text{K} \quad W = 500 \text{ g catalyst}$$

$$x = 0.50$$

$$r_c = 2.5 \frac{\text{g}}{\text{min}} \quad e = 0.4$$

$$a) \quad \frac{dF_A}{dV} = -\eta \kappa'' S_A r_c (1-\epsilon) C_A \quad F_A = C_A V$$

$$\frac{dC_A}{dV} = -\frac{\eta \kappa'' S_A r_c (1-\epsilon) C_A}{V}$$

$$\int \frac{dC_A}{C_A} = \int -\frac{\eta \kappa'' S_A r_c (1-\epsilon)}{V} dV \quad A = \frac{\eta \kappa'' S_A r_c (1-\epsilon)}{V}$$

$$\ln \left(\frac{C_A}{C_{A0}} \right) = -A W (1-\epsilon) r_c$$

$$\ln \left(\frac{\frac{1}{2} C_{A0}}{C_{A0}} \right) = -A W (1-\epsilon) r_c$$

$$\frac{-\ln \left(\frac{1}{2} \right)}{W (1-\epsilon) r_c} = A \quad A = 0.000924$$

$$b) \quad k_1'' = A e^{\frac{-E_a}{RT}}$$

$$1 \cdot 10^5 \frac{\text{cm}}{\text{m}} \cdot \frac{\ln}{100 \text{ cm}} \cdot \frac{1 \text{ yr}}{10 \text{ s}} = 11.67$$

$$k_1'' = 11.67 e^{\frac{-100,000}{8.314 \cdot 100}}$$

$$\frac{100 \text{ J}}{\text{mol}} \cdot \frac{1 \text{ mol}}{1 \text{ J}} = 100,000 \frac{\text{J}}{\text{mol}}$$

$$k_1'' = 1.45 \cdot 10^{-12} \frac{\text{m}}{\text{s}}$$

$$k_2'' = k_1 e^{\frac{-E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)}$$

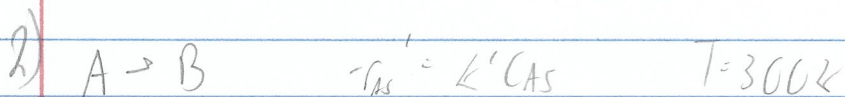
$$= 8.53 \cdot 10^{-12} \frac{\text{m}}{\text{s}}$$

$$A_2 = A_1 \frac{k_2'' (k_1'')^{\frac{1}{2}}}{k_1'' (k_2'')^{\frac{1}{2}}}$$

$$A_2 = 0.002237505$$

$$\frac{-\ln\left(\frac{1}{2}\right)}{A(1-\epsilon)r_2} = W$$

$$W = 201 \text{ g cetylrat}$$



$$\bar{V}_0 = \frac{16 \text{ cm}^3}{s}$$

$$Sh = 100 Re^{\frac{1}{2}}$$

$$\alpha = 0.10$$

$$C_{A0} = 1.0 \frac{\text{mol}}{L}$$

$$Re = \frac{9 \text{ cm} \cdot 15 \text{ cm}}{0.02 \text{ cm}} = 67.5$$

$$Sh = 821.6$$

$$Sh = \frac{k_c d_p}{D_{AB}}$$

$$\frac{Sh D_{AB}}{d_p} = k_c$$

$$k_c = 54.77 \frac{\text{cm}}{s}$$

$$\frac{\text{cm}}{s}$$

$$k_{eff} = \frac{k_c k''}{(k_c + k'')}$$

$$k'' = \frac{k'}{a} = \frac{0.01 \frac{\text{cm}^2}{s}}{0.02 \text{ cm}} = 0.00016667 \frac{\text{cm}}{s}$$

$$k_{eff} = 0.00017 \cdot 10^{-4} \frac{\text{cm}}{s}$$

$$\frac{dT_A}{dW} = k_{eff} C_A a$$

$$\ln \left(\frac{C_A}{C_{A0}} \right) = \frac{-W}{\bar{V}_0} k_{eff} a$$

$$C_A = (1 - \alpha) C_{A0} = 0.9 C_{A0}$$

$$\bar{V}_0 \ln \left(\frac{C_A}{C_{A0}} \right)$$

$$W = 9.16 \cdot 10^5 g$$

$$k_{eff} a$$

$$\frac{\text{cm}^2}{s} \cdot \frac{\text{cm}^2}{\text{cm}^2} = \frac{\text{cm}^2}{s}$$